

Linux OS and Scripting

Day-1 Assignment

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Linux Administration and Shell Commands

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1 Part A — User and System Investigation

1.1 1.1 Determine the Username of the Current Account

Use the following command:

```
whoami
```

The `whoami` command shows the username of the account currently in use.

1.2 1.2 Display the UID, Primary GID, and Supplementary Groups

Use:

```
id
```

This command displays the User ID (UID), the primary Group ID (GID), and any supplementary group memberships.

1.3 1.3 Display All Currently Logged-In Users

Use:

```
who
```

The command lists all users who are currently logged in to the system.

1.4 1.4 Determine the Activities of Logged-In Users

Use:

```
w
```

This command shows the logged-in users and information about what they are doing.

1.5 1.5 Examine the Current Account in `/etc/passwd`

Use:

```
grep "$(whoami):" /etc/passwd
```

The command output provides the following account details:

Information	Value Obtained
Username	ribhav
UID	1000
GID	1000
Home Directory	/home/ribhav
Login Shell	/bin/bash

1.6 1.6 Determine the Number of User-Account Entries

Use:

```
wc -l /etc/passwd
```

Each line in `/etc/passwd` represents one user account entry.

1.7 1.7 Display Only the Username and UID Fields

Use:

```
cut -d: -f1,3 /etc/passwd
```

The first field is the username, and the third field is the UID.

1.8 1.8 Find All Accounts with UID 0

Use:

```
awk -F: '$3 == 0 { print $1 }' /etc/passwd
```

UID 0 gives superuser privileges. Accounts with this UID have full administrative access, so they must be carefully protected.

1.9 1.9 Compare `/etc/passwd` and `/etc/shadow`

<code>/etc/passwd</code>	<code>/etc/shadow</code>
Stores general information about user accounts.	Stores password hashes and password-aging information.
Usually readable by regular users and system programs.	Restricted to privileged users such as root.
Contains the username, UID, primary GID, home directory, and login shell.	Contains encrypted password hashes and password-expiry details.
The password field normally contains x, showing that the hash is stored elsewhere.	Stores the protected password hash itself.

Password hashes are kept in `/etc/shadow` because `/etc/passwd` is normally readable by regular users. This separation protects the hashes and lowers the risk of unauthorized password-cracking attempts.

2 Part B — Creating and Verifying Users

2.1 2.1 Create the Required Users

Create the following users: alice, bob, and charlie.

```
sudo useradd -m alice
sudo useradd -m bob
sudo useradd -m charlie
```

The -m option automatically creates a home directory for each user.

2.2 2.2 Verify That the Accounts Were Added

Use:

```
grep -E '^(alice|bob|charlie):' /etc/passwd
```

This checks that all three accounts were added successfully.

2.3 2.3 Determine the UID Assigned to Each User

Use:

```
id -u alice
id -u bob
id -u charlie
```

The obtained UIDs are:

Username	UID
alice	1001
bob	1002
charlie	1003

2.4 2.4 Determine Alice's Home Directory and Default Shell

Use:

```
grep "^alice:" /etc/passwd
```

The relevant details are:

Information	Value Obtained
Home Directory	/home/alice
Default Shell	/bin/sh

2.5 2.5 Inspect the /home Directory

Use:

```
ls -l /home
```

The following directories should be present:

- /home/alice

- /home/bob
- /home/charlie

2.6 2.6 Set Passwords for All Three Accounts

Use:

```
sudo passwd alice
sudo passwd bob
sudo passwd charlie
```

Enter and confirm a password for each account.

2.7 2.7 Switch to Alice Using a Login Shell

Use:

```
su - alice
whoami
```

The output should be:

```
alice
```

2.8 2.8 Return to the Original Account

Use:

```
exit
whoami
```

The final command should display the username of the original account.

3 Part C — Group Management and Access Control

3.1 3.1 Create the Required Groups

Create the groups security and developers.

```
sudo groupadd security
sudo groupadd developers
```

3.2 3.2 Add Alice and Charlie to the Security Group

Use:

```
sudo usermod -aG security alice
sudo usermod -aG security charlie
```

These commands add both Alice and Charlie to the security group.

3.3 3.3 Add Bob to the Developers Group

Use:

```
sudo usermod -aG developers bob
```

This command adds Bob to the developers group.

3.4 3.4 Verify the Group Membership of All Users

Without changing accounts, use:

```
id alice
id bob
id charlie
```

Alternatively, use:

```
groups alice
groups bob
groups charlie
```

The expected supplementary groups are:

User	Supplementary Group
alice	security
bob	developers
charlie	security

3.5 3.5 Display the Security and Developers Group Entries

Use:

```
grep -E '^(security|developers):' /etc/group
```

This command displays the entries for the security and developers groups.

3.6 3.6 Add Bob to the Security Group

Use:

```
sudo usermod -aG security bob
```

The `-aG` option appends security to Bob's existing supplementary groups. Therefore, Bob remains a member of developers as well.

3.7 3.7 Verify That Bob Belongs to Both Groups

Use either command:

```
groups bob  
id bob
```

Bob's final group membership is:

User	Groups
bob	developers, security

End of Assignment